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Study Stuff for MCA Examinations

nimcet

Previous Year Questions



nimcet Previous Year Questions (PYQs)

nimcet 2015 PYQ

Qus : 1 **nimcet PYQ 2015** (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=year&key=2015>)

If $\vec{a} = 4\hat{j}$ and $\vec{b} = 3\hat{j} + 4\hat{k}$, then the vector form of the component of $\vec{a}$ along $\vec{b}$ is

1

$$\frac{18}{10\sqrt{3}}(3\hat{j} + 4\hat{k})$$

2

$$\frac{18}{5}(3\hat{j} + 4\hat{k})$$

3

$$\frac{18}{10\sqrt{13}}(3\hat{j} + 4\hat{k})$$

4

$$(3\hat{j} + 4\hat{k})$$

[Go to Discussion \(https://www.test.aspirestudy.in/library/qus/10240\)](https://www.test.aspirestudy.in/library/qus/10240)

nimcet Previous Year PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=sub&key=Previous Year>)

nimcet NIMCET 2015 PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=topic&key=NIMCET>)

2015)

Qus : 2 nimcet PYQ 2015 (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=year&key=2015>)

If $\vec{a}$ and $\vec{b}$ in space, given by $\vec{a} = \frac{\hat{i}-2\hat{j}}{\sqrt{5}}$ and $\vec{b} = \frac{2\hat{i}+\hat{j}+3\hat{k}}{\sqrt{14}}$, then the value of $(2\vec{a} + \vec{b}) \cdot [(\vec{a} \times \vec{b}) \times (\vec{a} - 2\vec{b})]$ is

1

3

2

4

3

5

4

6

[Go to Discussion \(https://www.test.aspirestudy.in/library/qus/10241\)](https://www.test.aspirestudy.in/library/qus/10241)

nimcet Previous Year PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=sub&key=Previous Year>)

nimcet NIMCET 2015 PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=topic&key=NIMCET 2015>)

Qus : 3 nimcet PYQ 2015 (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=year&key=2015>)

The letters of English alphabet from A to M were written, leaving space for one letter between every two letters and then the remaining letters were inserted beginning with N and ending the series with Z after M. Which letter would be 3rd to the right of the 7th letter from the left

1

C

2

O

3

R

4

S

[Go to Discussion \(https://www.test.aspirestudy.in/library/qus/10242\)](https://www.test.aspirestudy.in/library/qus/10242)

nimcet Previous Year PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=sub&key=Previous Year>)

nimcet NIMCET 2015 PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=topic&key=NIMCET 2015>)

Qus : 4 nimcet PYQ 2015 (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=year&key=2015>)

The letters of English alphabet from A to M were written, leaving space for one letter between every two letters and then the remaining letters were inserted beginning with N and ending the series with Z after M. Which letter would be exactly in the middle of eighteenth letter from the beginning and fifteenth from the end?

- ☐ 1 G
- ☐ 2 H
- ☐ 3 J
- ☐ 4 L

[Go to Discussion \(https://www.test.aspirestudy.in/library/qus/10243\)](https://www.test.aspirestudy.in/library/qus/10243)

nimcet Previous Year PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=sub&key=Previous Year>)

nimcet NIMCET 2015 PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=topic&key=NIMCET 2015>)

Qus : 5 nimcet PYQ 2015 (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=year&key=2015>)

How many 3-digit numbers divisible by 5, can be formed using the digits 2 3 5 6 7 and 9, without repetition of digits?

- ☐ 1 216
- ☐ 2 20
- ☐ 3 120
- ☐ 4 24

[Go to Discussion \(https://www.test.aspirestudy.in/library/qus/10244\)](https://www.test.aspirestudy.in/library/qus/10244)

nimcet Previous Year PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=sub&key=Previous Year>)

nimcet NIMCET 2015 PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=topic&key=NIMCET 2015>)

Qus : 6 nimcet PYQ 2015 (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=year&key=2015>)

Union sets A, B, C, D, E and F are given below, what is the smallest number of sets required to represent the universal set?

Using only 2, 5, 10, 25 and 50 paise coins, what is the smallest number of coins required to pay exactly 78 paise, 69 paise and Rs. 1.01 to three different persons?

- 1 19
- 2 20
- 3 17
- 4 18

Go to Discussion (<https://www.test.aspirestudy.in/library/qus/10245>)

nimcet Previous Year PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=sub&key=Previous Year>)

nimcet NIMCET 2015 PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=topic&key=NIMCET 2015>)

Qus : 7 nimcet PYQ 2015 (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=year&key=2015>)

Which of the following two patterns will fit in the blanks of the series $ZA_5, Y_4B, XC_6, W_3D, \underline{\hspace{2cm}}, \underline{\hspace{2cm}}?$

- 1 VE_7 and U_2E
- 2 V_2E and U_7F
- 3 VE_7 and U_2F
- 4 VF_7 and U_2E

Go to Discussion (<https://www.test.aspirestudy.in/library/qus/10246>)

nimcet Previous Year PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=sub&key=Previous Year>)

nimcet NIMCET 2015 PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=topic&key=NIMCET 2015>)

Qus : 8 nimcet PYQ 2015 (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=year&key=2015>)

Which of the following numbers comes next in the two-digit decimal number sequence 61, 52, 63, 94, $\underline{\hspace{2cm}}?$

- 1 65

2

64

3

56

4

46

[Go to Discussion \(https://www.test.aspirestudy.in/library/qus/10247\)](https://www.test.aspirestudy.in/library/qus/10247)

nimcet Previous Year PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=sub&key=Previous Year>)

nimcet NIMCET 2015 PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=topic&key=NIMCET 2015>)

Qus : 9 nimcet PYQ 2015 (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=year&key=2015>)

Three ladies X, Y and Z marry three men A, B and C. X is married to A, Y is not married to an engineer. Z is not married to a doctor, C is not a doctor and A is a lawyer. Then which of the following statements is correct?

1

X is married to a doctor

2

Y is married to C, who is a doctor

3

Z is married to B, who is an engineer

4

None of these

[Go to Discussion \(https://www.test.aspirestudy.in/library/qus/10248\)](https://www.test.aspirestudy.in/library/qus/10248)

nimcet Previous Year PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=sub&key=Previous Year>)

nimcet NIMCET 2015 PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=topic&key=NIMCET 2015>)

Qus : 10 nimcet PYQ 2015 (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=year&key=2015>)

There are five books A, B, C, D and E placed on table. If A is placed below E, C is placed above D, B is placed below A and D is placed between A and E, then which of the following books can be on the top?

1

D or E

2

C or E

3

A or E

4

None of these

[Go to Discussion \(https://www.test.aspirestudy.in/library/qus/10249\)](https://www.test.aspirestudy.in/library/qus/10249)

nimcet Previous Year PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=sub&key=Previous Year>)

nimcet NIMCET 2015 PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=topic&key=NIMCET 2015>)

Qus : 11 nimcet PYQ 2015 (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=year&key=2015>)

Among five children A, B, C, D and E, B is taller than E but shorter than D. A is shorter than C but taller than D. If all the children stand in a line according to their heights, then who would be the fourth if counted from the tallest one?

1

D

2

C

3

B

4

A

[Go to Discussion \(https://www.test.aspirestudy.in/library/qus/10250\)](https://www.test.aspirestudy.in/library/qus/10250)

nimcet Previous Year PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=sub&key=Previous Year>)

nimcet NIMCET 2015 PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=topic&key=NIMCET 2015>)

Qus : 12 nimcet PYQ 2015 (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=year&key=2015>)

- In a family of six-person A, B, C, D, E, and F there are two married couples.
- D is the grandmother of A and mother of B.
- C is the wife of B and mother of F
- F is the granddaughter of E

What is C to A?

1

Daughter

2

Grandmother

3

Mother

4

Cannot be determined

[Go to Discussion \(https://www.test.aspirestudy.in/library/qus/10251\)](https://www.test.aspirestudy.in/library/qus/10251)

nimcet Previous Year PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=sub&key=Previous Year>)

nimcet NIMCET 2015 PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=topic&key=NIMCET 2015>)

Qus : 13 nimcet PYQ 2015 (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=year&key=2015>)

- In a family of six-person A, B, C, D, E, and F there are two married couples.
- D is the grandmother of A and mother of B.
- C is the wife of B and mother of F
- F is the granddaughter of E

How many male members are there in the family?

1

Two

2

Three

3

Four

4

Cannot be determined

[Go to Discussion \(https://www.test.aspirestudy.in/library/qus/10252\)](https://www.test.aspirestudy.in/library/qus/10252)

nimcet Previous Year PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=sub&key=Previous Year>)

nimcet NIMCET 2015 PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=topic&key=NIMCET 2015>)

Qus : 14 nimcet PYQ 2015 (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=year&key=2015>)

- In a family of six-person A, B, C, D, E, and F there are two married couples.
- D is the grandmother of A and mother of B.
- C is the wife of B and mother of F
- F is the granddaughter of E

Who among the following is one of the couples?

- 1 CD
- 2 DE
- 3 EB
- 4 Cannot be determined

[Go to Discussion \(https://www.test.aspirestudy.in/library/qus/10253\)](https://www.test.aspirestudy.in/library/qus/10253)

nimcet Previous Year PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=sub&key=Previous Year>)

nimcet NIMCET 2015 PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=topic&key=NIMCET 2015>)

Qus : 15 nimcet PYQ 2015 (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=year&key=2015>)

- In a family of six-person A, B, C, D, E, and F there are two married couples.
- D is the grandmother of A and mother of B.
- C is the wife of B and mother of F
- F is the granddaughter of E

Which of the following is true?

- 1 A is brother of F
- 2 A is sister of F
- 3 B has two daughters
- 4 None of these

[Go to Discussion \(https://www.test.aspirestudy.in/library/qus/10254\)](https://www.test.aspirestudy.in/library/qus/10254)

nimcet Previous Year PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=sub&key=Previous Year>)

nimcet NIMCET 2015 PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=topic&key=NIMCET 2015>)

Qus : 16 nimcet PYQ 2015 (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=year&key=2015>)

A, B, C, D, E, F and G are seven girls having different amount of money from among Rs. 10, 20, 40, 60, 80, 120 and 200 with them. They had 3 chocolates, 2 toffees, and 2 lollipops together, each one having one of these seven items.

- B and F do not have chocolates and they have Rs. 200 and Rs. 80 respectively.

- C has Rs. 60 with her and G has an amount which is neither Rs. 40 nor Rs. 120.
- A has Rs. 10 and does not have a toffee.
- The girl having Rs. 40 with her is the only one other than A to have the same type of item.
- E and the girl having Rs. 20 with her have the same kind of item.

How much amount does G have with her?

- 1 Rs. 20
- 2 Rs. 10
- 3 Rs. 60
- 4 None of these

[Go to Discussion \(https://www.test.aspirestudy.in/library/qus/10255\)](https://www.test.aspirestudy.in/library/qus/10255)

nimcet Previous Year PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=sub&key=Previous Year>)

nimcet NIMCET 2015 PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=topic&key=NIMCET 2015>)

Qus : 17 nimcet PYQ 2015 (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=year&key=2015>)

A, B, C, D, E, F and G are seven girls having different amount of money from among Rs. 10, 20, 40, 60, 80, 120 and 200 with them. They had 3 chocolates, 2 toffees, and 2 lollipops together, each one having one of these seven items.

- B and F do not have chocolates and they have Rs. 200 and Rs. 80 respectively.
- C has Rs. 60 with her and G has an amount which is neither Rs. 40 nor Rs. 120.
- A has Rs. 10 and does not have a toffee.
- The girl having Rs. 40 with her is the only one other than A to have the same type of item.
- E and the girl having Rs. 20 with her have the same kind of item.

Which of the following girls have chocolates with them?

- 1 F, C, G
- 2 C, G, E
- 3 C, G, D
- 4 G, D, E

[Go to Discussion \(https://www.test.aspirestudy.in/library/qus/10256\)](https://www.test.aspirestudy.in/library/qus/10256)

nimcet Previous Year PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=sub&key=Previous Year>)

Year)

nimcet NIMCET 2015 PYQ ([https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=topic&key=NIMCET 2015](https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=topic&key=NIMCET2015))

Qus : 18 nimcet PYQ 2015 (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=year&key=2015>)

A, B, C, D, E, F and G are seven girls having different amount of money from among Rs. 10, 20, 40, 60, 80, 120 and 200 with them. They had 3 chocolates, 2 toffees, and 2 lollipops together, each one having one of these seven items.

- B and F do not have chocolates and they have Rs. 200 and Rs. 80 respectively.
- C has Rs. 60 with her and G has an amount which is neither Rs. 40 nor Rs. 120.
- A has Rs. 10 and does not have a toffee.
- The girl having Rs. 40 with her is the only one other than A to have the same type of item.
- E and the girl having Rs. 20 with her have the same kind of item.

Which of the following combination is definitely correct?

1

C – chocolate – Rs. 60

2

G – toffee – Rs. 20

3

D – chocolate – Rs. 40

4

None of these

[Go to Discussion \(https://www.test.aspirestudy.in/library/qus/10257\)](https://www.test.aspirestudy.in/library/qus/10257)

nimcet Previous Year PYQ ([https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=sub&key=Previous Year](https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=sub&key=PreviousYear))

nimcet NIMCET 2015 PYQ ([https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=topic&key=NIMCET 2015](https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=topic&key=NIMCET2015))

Qus : 19 nimcet PYQ 2015 (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=year&key=2015>)

A, B, C, D, E, F and G are seven girls having different amount of money from among Rs. 10, 20, 40, 60, 80, 120 and 200 with them. They had 3 chocolates, 2 toffees, and 2 lollipops together, each one having one of these seven items.

- B and F do not have chocolates and they have Rs. 200 and Rs. 80 respectively.
- C has Rs. 60 with her and G has an amount which is neither Rs. 40 nor Rs. 120.
- A has Rs. 10 and does not have a toffee.
- The girl having Rs. 40 with her is the only one other than A to have the same type of item.
- E and the girl having Rs. 20 with her have the same kind of item.

Which girl has Rs. 40 with her?

1

–

1 E

2 A

3 D

4 None of these

[Go to Discussion \(https://www.test.aspirestudy.in/library/qus/10258\)](https://www.test.aspirestudy.in/library/qus/10258)

nimcet Previous Year PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=sub&key=Previous Year>)

nimcet NIMCET 2015 PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=topic&key=NIMCET 2015>)

Qus : 20 nimcet PYQ 2015 (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=year&key=2015>)

P, Q, R, S, T, U and V are sitting in a row facing North. In order to determine, who is sitting exactly in the middle of the row, which of the following information is needed?

- T and U are sitting at extreme ends of the row
- S is third to the right of T
- Q is four places to the left of R and P is two places to the left of V

1 I and II only are sufficient

2 I and III only are sufficient

3 I and either II or III are sufficient

4 I, II and III

[Go to Discussion \(https://www.test.aspirestudy.in/library/qus/10259\)](https://www.test.aspirestudy.in/library/qus/10259)

nimcet Previous Year PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=sub&key=Previous Year>)

nimcet NIMCET 2015 PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=topic&key=NIMCET 2015>)

Qus : 21 nimcet PYQ 2015 (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=year&key=2015>)

How many times do the hour and the minute hands of a clock overlap in 24 hours?

1 24

2

22

3

26

4

20

[Go to Discussion \(https://www.test.aspirestudy.in/library/qus/10260\)](https://www.test.aspirestudy.in/library/qus/10260)

nimcet Previous Year PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=sub&key=Previous Year>)

nimcet NIMCET 2015 PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=topic&key=NIMCET 2015>)

Qus : 22 nimcet PYQ 2015 (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=year&key=2015>)

In a certain code, TOGETHER is coded as RQEGRJCT. In the same code, PAROLE will be written as:

1

NCPQJG

2

NCQPJG

3

RCPQJK

4

RCTQNG

[Go to Discussion \(https://www.test.aspirestudy.in/library/qus/10261\)](https://www.test.aspirestudy.in/library/qus/10261)

nimcet Previous Year PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=sub&key=Previous Year>)

nimcet NIMCET 2015 PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=topic&key=NIMCET 2015>)

Qus : 23 nimcet PYQ 2015 (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=year&key=2015>)

A drawer contains 10 black and 10 brown socks which are all mixed up. What is the smallest number of socks to be taken from the drawer to decide without seeing them, to be sure that there is atleast one pair of socks of the same colour?

1

11

2

10

3

3

4

Cannot be determined

[Go to Discussion \(https://www.test.aspirestudy.in/library/qus/10262\)](https://www.test.aspirestudy.in/library/qus/10262)

nimcet Previous Year PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=sub&key=Previous Year>)

nimcet NIMCET 2015 PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=topic&key=NIMCET 2015>)

Qus : 24 nimcet PYQ 2015 (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=year&key=2015>)

Find the missing number in the series: 4, 7, 25, 10, _____, 20, 16, 19

1

13

2

15

3

20

4

28

[Go to Discussion \(https://www.test.aspirestudy.in/library/qus/10263\)](https://www.test.aspirestudy.in/library/qus/10263)

nimcet Previous Year PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=sub&key=Previous Year>)

nimcet NIMCET 2015 PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=topic&key=NIMCET 2015>)

Qus : 25 nimcet PYQ 2015 (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=year&key=2015>)

A circular field with inner radius of 10 meters and outer radius of 20 meters is divided into 5 successive stages for ploughing, The ploughing at each stage, with starting points P1, P2, P3, P4 and P5, was allotted to one of the five farmers F1, F2, F3, F4 and F5, not necessarily in that order.

- F was allotted the stage starting at point P4.
- The stage from P5 to P3 was not the first Stage.
- F4 was allotted the work of the fourth stage.
- Finishing point of stage 3 was P1 and the work was not allotted to F1.
- F3 was allotted the work of stage 3 was P1 and the work was not allotted to F1.

Which of the following is the finish point for farmer F2?

1

P1

2

P2

3

P3

4

P4

[Go to Discussion \(https://www.test.aspirestudy.in/library/qus/10264\)](https://www.test.aspirestudy.in/library/qus/10264)

nimcet Previous Year PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=sub&key=Previous Year>)

nimcet NIMCET 2015 PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=topic&key=NIMCET 2015>)

Qus : 26 nimcet PYQ 2015 (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=year&key=2015>)

A circular field with inner radius of 10 meters and outer radius of 20 meters is divided into 5 successive stages for ploughing, The ploughing at each stage, with starting points P1, P2, P3, P4 and P5, was allotted to one of the five farmers F1, F2, F3, F4 and F5, not necessarily in that order.

- F was allotted the stage starting at point P4.
- The stage from P5 to P3 was not the first Stage.
- F4 was allotted the work of the fourth stage.
- Finishing point of stage 3 was P1 and the work was not allotted to F1.
- F3 was allotted the work of stage 3 was P1 and the work was not allotted to F1.

Which stage was ploughed by F5?

1

2

2

3

3

4

4

5

[Go to Discussion \(https://www.test.aspirestudy.in/library/qus/10265\)](https://www.test.aspirestudy.in/library/qus/10265)

nimcet Previous Year PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=sub&key=Previous Year>)

nimcet NIMCET 2015 PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=topic&key=NIMCET 2015>)

Qus : 27 nimcet PYQ 2015 (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=year&key=2015>)

A circular field with inner radius of 10 meters and outer radius of 20 meters is divided into 5 successive stages for ploughing, The ploughing at each stage, with starting points P1, P2, P3, P4 and P5, was allotted

to one of the five farmers F1, F2, F3, F4 and F5, not necessarily in that order.

- F was allotted the stage starting at point P4.
- The stage from P5 to P3 was not the first Stage.
- F4 was allotted the work of the fourth stage.
- Finishing point of stage 3 was P1 and the work was not allotted to F1.
- F3 was allotted the work of stage 3 was P1 and the work was not allotted to F1.

What are the starting and ending points of the field ploughed by F4?

- 1 P1 and P2
- 2 P1 and P4
- 3 P4 and P2
- 4 P2 and P4

[Go to Discussion \(https://www.test.aspirestudy.in/library/qus/10266\)](https://www.test.aspirestudy.in/library/qus/10266)

nimcet Previous Year PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=sub&key=Previous Year>)

nimcet NIMCET 2015 PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=topic&key=NIMCET 2015>)

Qus : 28 nimcet PYQ 2015 (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=year&key=2015>)

A circular field with inner radius of 10 meters and outer radius of 20 meters is divided into 5 successive stages for ploughing, The ploughing at each stage, with starting points P1, P2, P3, P4 and P5, was allotted to one of the five farmers F1, F2, F3, F4 and F5, not necessarily in that order.

- F was allotted the stage starting at point P4.
- The stage from P5 to P3 was not the first Stage.
- F4 was allotted the work of the fourth stage.
- Finishing point of stage 3 was P1 and the work was not allotted to F1.
- F3 was allotted the work of stage 3 was P1 and the work was not allotted to F1.

What is the starting point for stage?

- 1 P2
- 2 P3
- 3 P4
- 4 Caanot be determined

[Go to Discussion \(https://www.test.aspirestudy.in/library/qus/10267\)](https://www.test.aspirestudy.in/library/qus/10267)

nimcet Previous Year PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=sub&key=Previous Year>)

nimcet NIMCET 2015 PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=topic&key=NIMCET 2015>)

Qus : 29 nimcet PYQ 2015 (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=year&key=2015>)

If Tuesday falls on the fourth of a month then which day will fall three days after 24th of the same month?

- ☒ 1 Monday
- ☐ 2 Tuesday
- ☐ 3 Thursday
- ☐ 4 Friday

[Go to Discussion \(https://www.test.aspirestudy.in/library/qus/10268\)](https://www.test.aspirestudy.in/library/qus/10268)

nimcet Previous Year PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=sub&key=Previous Year>)

nimcet NIMCET 2015 PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=topic&key=NIMCET 2015>)

Qus : 30 nimcet PYQ 2015 (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=year&key=2015>)

If the statements “All chickens are birds”, “Some chickens are hens” and “Female birds lay eggs”, are all facts, then which of the following must also be a fact?

- I. All birds lay eggs
- II. Some hens are birds
- III. Some chickens are not hens

- ☒ 1 I and II
- ☐ 2 II and III
- ☐ 3 I and III
- ☐ 4 Neither I nor II nor III

[Go to Discussion \(https://www.test.aspirestudy.in/library/qus/10269\)](https://www.test.aspirestudy.in/library/qus/10269)

nimcet Previous Year PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=sub&key=Previous Year>)

nimcet NIMCET 2015 PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=topic&key=NIMCET 2015>)

Qus : 31 nimcet PYQ 2015 (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=year&key=2015>)

- There is a family of six members A, B, C, D, E and F
- There are two married couples in the family and the family members represent three generations.
- Each member has a distinct choice of a colour amongst Green, Yellow, Black, Red, White and Pink
- No lady member likes either Green or White.
- C, who likes Black colour, is the daughter-in-law of E.
- B is the brother of F and son of D and likes Pink
- A is the grandmother of F and F does not like Red.
- Wife of the husband having a choice for Green colour, likes Yellow.

Which of the following is the colour preference of A?

- 1 Red
- 2 Yellow
- 3 Either Red or Yellow
- 4 Cannot be determined

[Go to Discussion \(https://www.test.aspirestudy.in/library/qus/10270\)](https://www.test.aspirestudy.in/library/qus/10270)

nimcet Previous Year PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=sub&key=Previous Year>)

nimcet NIMCET 2015 PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=topic&key=NIMCET 2015>)

Qus : 32 nimcet PYQ 2015 (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=year&key=2015>)

- There is a family of six members A, B, C, D, E and F
- There are two married couples in the family and the family members represent three generations.
- Each member has a distinct choice of a colour amongst Green, Yellow, Black, Red, White and Pink
- No lady member likes either Green or White.
- C, who likes Black colour, is the daughter-in-law of E.
- B is the brother of F and son of D and likes Pink
- A is the grandmother of F and F does not like Red.
- Wife of the husband having a choice for Green colour, likes Yellow.

Which of the following could be the colour combination of one of the couples?

1

Yellow-Red

2

Green-Black

3

Red-Yellow

4

Yellow-Green

[Go to Discussion \(https://www.test.aspirestudy.in/library/qus/10271\)](https://www.test.aspirestudy.in/library/qus/10271)

nimcet Previous Year PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=sub&key=Previous Year>)

nimcet NIMCET 2015 PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=topic&key=NIMCET 2015>)

Qus : 33 nimcet PYQ 2015 (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=year&key=2015>)

- There is a family of six members A, B, C, D, E and F
- There are two married couples in the family and the family members represent three generations.
- Each member has a distinct choice of a colour amongst Green, Yellow, Black, Red, White and Pink
- No lady member likes either Green or White.
- C, who likes Black colour, is the daughter-in-law of E.
- B is the brother of F and son of D and likes Pink
- A is the grandmother of F and F does not like Red.
- Wife of the husband having a choice for Green colour, likes Yellow.

Which of the following is one of the married couples?

1

CD

2

AC

3

AD

4

Cannot be determined

[Go to Discussion \(https://www.test.aspirestudy.in/library/qus/10272\)](https://www.test.aspirestudy.in/library/qus/10272)

nimcet Previous Year PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=sub&key=Previous Year>)

nimcet NIMCET 2015 PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=topic&key=NIMCET 2015>)

Qus : 34 nimcet PYQ 2015 (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=year&key=2015>)

- There is a family of six members A, B, C, D, E and F
- There are two married couples in the family and the family members represent three generations.
- Each member has a distinct choice of a colour amongst Green, Yellow, Black, Red, White and Pink
- No lady member likes either Green or White.
- C, who likes Black colour, is the daughter-in-law of E.
- B is the brother of F and son of D and likes Pink
- A is the grandmother of F and F does not like Red.
- Wife of the husband having a choice for Green colour, likes Yellow.

Which of the following is true about F?

- 1 Brother of B
- 2 Sister of B
- 3 Daughter of C
- 4 Cannot be determined

[Go to Discussion \(https://www.test.aspirestudy.in/library/qus/10273\)](https://www.test.aspirestudy.in/library/qus/10273)

nimcet Previous Year PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=sub&key=Previous Year>)

nimcet NIMCET 2015 PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=topic&key=NIMCET 2015>)

Qus : 35 nimcet PYQ 2015 (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=year&key=2015>)

If the English word “EXAMINATION” is coded as 56149512965, then the word “GOVERNMENT” is coded as

- 1 7645954552
- 2 7654694562
- 3 7645955423
- 4 7654964526

[Go to Discussion \(https://www.test.aspirestudy.in/library/qus/10274\)](https://www.test.aspirestudy.in/library/qus/10274)

nimcet Previous Year PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=sub&key=Previous Year>)

nimcet NIMCET 2015 PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=topic&key=NIMCET 2015>)

Qus : 36 nimcet PYQ 2015 (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=year&key=2015>)

Gopal starts from his house towards West. After walking a distance of 30 meters, he turned towards right and walked 20 meters. He turned left and after moving a distance of 10 meters, turned to his left again and walked 40 meters. He then turned left and walked 5 meters. Finally, he turns to his left. In which direction is he walking now?

- 1 North
- 2 South
- 3 East
- 4 South West

[Go to Discussion \(https://www.test.aspirestudy.in/library/qus/10275\)](https://www.test.aspirestudy.in/library/qus/10275)

nimcet Previous Year PYQ ([https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=sub&key=Previous Year](https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=sub&key=Previous%20Year))

nimcet NIMCET 2015 PYQ ([https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=topic&key=NIMCET 2015](https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=topic&key=NIMCET%2015))

Qus : 37 nimcet PYQ 2015 (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=year&key=2015>)

Read the conclusion and then decide which of the given conclusions logically follows from the two given statements, (i) and (ii) disregarding commonly known facts.

Statements:

- (i) No woman teacher can play.
- (ii) Some woman teachers are athletes.

Conclusions:

- I. Male athletes can play.
- II. Some athletes can play

- 1 Only conclusion I follows.
- 2 Only conclusion II follows
- 3 Either I or II follows
- 4 Neither I nor II follows

[Go to Discussion \(https://www.test.aspirestudy.in/library/qus/10276\)](https://www.test.aspirestudy.in/library/qus/10276)

nimcet Previous Year PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=sub&key=Previous Year>)

nimcet NIMCET 2015 PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=topic&key=NIMCET 2015>)

Qus : 38 nimcet PYQ 2015 (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=year&key=2015>)

Which of the following numbers come next in the series 8, 6, 9, 23, 87, _____?

1

128

2

226

3

324

4

429

[Go to Discussion \(https://www.test.aspirestudy.in/library/qus/10277\)](https://www.test.aspirestudy.in/library/qus/10277)

nimcet Previous Year PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=sub&key=Previous Year>)

nimcet NIMCET 2015 PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=topic&key=NIMCET 2015>)

Qus : 39 nimcet PYQ 2015 (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=year&key=2015>)

In an examination, there are 100 questions divided into 3 parts A, B, C, and each part should contain at least one question. Each question in parts A, B, and C carry 1, 2 and 3 marks respectively. Part A is for at least 60% of the total marks and part B should contain 23 questions. How many questions must be set in part C?

1

1

2

2

3

3

4

Cannot be determined.

[Go to Discussion \(https://www.test.aspirestudy.in/library/qus/10278\)](https://www.test.aspirestudy.in/library/qus/10278)

nimcet Previous Year PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=sub&key=Previous Year>)

year)

nimcet NIMCET 2015 PYQ ([https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=topic&key=NIMCET 2015](https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=topic&key=NIMCET2015))

Qus : 40 nimcet PYQ 2015 (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=year&key=2015>)

If $\div$ means addition, $-$ means division, $\times$ means subtraction and $+$ means multiplication, then the value of $\frac{(36 \times 4) - (8 \times 4)}{4 + 8 \times 2 + 16 \div 1}$

- ☐ 1
- ☐ 2
- ☐ 3
- ☐ 4

[Go to Discussion \(https://www.test.aspirestudy.in/library/qus/10279\)](https://www.test.aspirestudy.in/library/qus/10279)

nimcet Previous Year PYQ ([https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=sub&key=Previous Year](https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=sub&key=PreviousYear))

nimcet NIMCET 2015 PYQ ([https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=topic&key=NIMCET 2015](https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=topic&key=NIMCET2015))

Qus : 41 nimcet PYQ 2015 (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=year&key=2015>)

Which letter in the word CYBERNETICS occupies the same position as it does in the English alphabet?

- ☐ 1
- ☐ 2
- ☐ 3
- ☐ 4

[Go to Discussion \(https://www.test.aspirestudy.in/library/qus/10280\)](https://www.test.aspirestudy.in/library/qus/10280)

nimcet Previous Year PYQ ([https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=sub&key=Previous Year](https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=sub&key=PreviousYear))

nimcet NIMCET 2015 PYQ ([https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=topic&key=NIMCET 2015](https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=topic&key=NIMCET2015))

Qus : 42 nimcet PYQ 2015 (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=year&key=2015>)

The remainder when 2^{31} is divided by 5 is

- 1
- 2
- 3
- 4

[Go to Discussion \(https://www.test.aspirestudy.in/library/qus/10281\)](https://www.test.aspirestudy.in/library/qus/10281)

nimcet Previous Year PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=sub&key=Previous Year>)

nimcet NIMCET 2015 PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=topic&key=NIMCET 2015>)

Qus : 43 nimcet PYQ 2015 (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=year&key=2015>)

Let $\vec{a}$ and $\vec{b}$ be two vectors, which of the following vectors are not perpendicular to each other?

- 1
- 2
- 3
- 4

[Go to Discussion \(https://www.test.aspirestudy.in/library/qus/10191\)](https://www.test.aspirestudy.in/library/qus/10191)

nimcet Previous Year PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=sub&key=Previous Year>)

nimcet NIMCET 2015 PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=topic&key=NIMCET 2015>)

Qus : 44 nimcet PYQ 2015 (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=year&key=2015>)

If $A = \begin{bmatrix} a & b & c \\ b & c & a \\ c & a & b \end{bmatrix}$, where a, b, c are real positive numbers such that $abc = 1$ and $A^T A = I$

then the equation that **not** holds true among the following is

then the equation that not holds true among the following is

1

$$a + b + c = 1$$

2

$$a^2 + b^2 + c^2 = 1$$

3

$$ab + bc + ca = 0$$

4

$$a^3 + b^3 + c^3 = 4$$

[Go to Discussion \(https://www.test.aspirestudy.in/library/qus/10193\)](https://www.test.aspirestudy.in/library/qus/10193)

nimcet Previous Year PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=sub&key=Previous Year>)

nimcet NIMCET 2015 PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=topic&key=NIMCET 2015>)

Qus : 45 nimcet PYQ 2015 (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=year&key=2015>)

The equation of the tangent at any point of the curve $x = a \cos 2t$, $y = 2\sqrt{2}a \sin t$ with m as its slope is

1

$$y = mx + a \left(m - \frac{1}{m} \right)$$

2

$$y = mx - a \left(m + \frac{1}{m} \right)$$

3

$$y = mx + m \left(a + \frac{1}{a} \right)$$

4

$$y = amx + a \left(m - \frac{1}{m} \right)$$

[Go to Discussion \(https://www.test.aspirestudy.in/library/qus/10194\)](https://www.test.aspirestudy.in/library/qus/10194)

nimcet Previous Year PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=sub&key=Previous Year>)

nimcet NIMCET 2015 PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=topic&key=NIMCET 2015>)

Qus : 46 nimcet PYQ 2015 (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=year&key=2015>)

The locus of the mid points of all chords of the parabola $y^2 = 4x$ which are drawn through its vertex, is

1

$$y^2 = 8x$$

2

$$y^2 = 2x$$

3

$$x^2 + 4y^2 = 16$$

4

$$x^2 = 2y$$

[Go to Discussion \(https://www.test.aspirestudy.in/library/qus/10195\)](https://www.test.aspirestudy.in/library/qus/10195)

nimcet Previous Year PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=sub&key=Previous Year>)

nimcet NIMCET 2015 PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=topic&key=NIMCET 2015>)

Qus : 47 nimcet PYQ 2015 (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=year&key=2015>)

The value of $\lim_{x \rightarrow a} \frac{\sqrt{a+2x} - \sqrt{3x}}{\sqrt{3a+x} - 2\sqrt{x}}$

1

$$\frac{2}{3}$$

2

$$\frac{2}{\sqrt{3}}$$

3

$$\frac{3\sqrt{3}}{\sqrt{2}}$$

4

$$\frac{2}{3\sqrt{3}}$$

[Go to Discussion \(https://www.test.aspirestudy.in/library/qus/10196\)](https://www.test.aspirestudy.in/library/qus/10196)

nimcet Previous Year PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=sub&key=Previous Year>)

Year)

nimcet NIMCET 2015 PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=topic&key=NIMCET> 2015)

Qus : 48 nimcet PYQ 2015 (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=year&key=2015>)

The value of $\int_{-\pi/3}^{\pi/3} \frac{x \sin x}{\cos^2 x} dx$

1

$$\frac{1}{3}(4\pi + 1)$$

2

$$\frac{4\pi}{3} - 2\log \tan \frac{5\pi}{12}$$

3

$$\frac{4\pi}{3} + \log \tan \frac{5\pi}{12}$$

4

$$\frac{4\pi}{3} - \log \tan \frac{5\pi}{12}$$

[Go to Discussion \(https://www.test.aspirestudy.in/library/qus/10197\)](https://www.test.aspirestudy.in/library/qus/10197)

nimcet Previous Year PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=sub&key=Previous> Year)

nimcet NIMCET 2015 PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=topic&key=NIMCET> 2015)

Qus : 49 nimcet PYQ 2015 (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=year&key=2015>)

The foci of the ellipse $\frac{x^2}{16} + \frac{y^2}{b^2} = 1$ and the hyperbola $\frac{x^2}{144} - \frac{y^2}{81} = \frac{1}{25}$ coincide, then the value of b^2 is

1

1

2

5

3

7

4

9

[Go to Discussion \(https://www.test.aspirestudy.in/library/qus/10198\)](https://www.test.aspirestudy.in/library/qus/10198)

nimcet Previous Year PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=sub&key=Previous Year>)

nimcet NIMCET 2015 PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=topic&key=NIMCET 2015>)

Qus : 50 nimcet PYQ 2015 (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=year&key=2015>)

If $a + b + c = \pi$, then the value of $\begin{vmatrix} \sin(A + B + C) & \sin B & \cos C \\ -\sin B & 0 & \tan A \\ \cos(A + B) & -\tan A & 0 \end{vmatrix}$ is

1

0

2

1

3

$2\sin A \sin B$

4

2

[Go to Discussion \(https://www.test.aspirestudy.in/library/qus/10199\)](https://www.test.aspirestudy.in/library/qus/10199)

nimcet Previous Year PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=sub&key=Previous Year>)

nimcet NIMCET 2015 PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=topic&key=NIMCET 2015>)

Qus : 51 nimcet PYQ 2015 (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=year&key=2015>)

If the mean deviation of the numbers $1, 1 + d, 1 + 2d, \dots, 1 + 100d$ from their mean is 255, then the value of d is

1

20.0

2

10.1

3

20.2

4

10.0

[Go to Discussion \(https://www.test.aspirestudy.in/library/qus/10200\)](https://www.test.aspirestudy.in/library/qus/10200)

nimcet Previous Year PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=sub&key=Previous Year>)

nimcet NIMCET 2015 PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=topic&key=NIMCET 2015>)

Qus : 52 nimcet PYQ 2015 (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=year&key=2015>)

If $P = \sin^{20}\theta + \cos^{48}\theta$ then the inequality that holds for all values of θ is

- 1 $P \geq 1$
- 2 $0 < P \leq 1$
- 3 $1 < P < 3$
- 4 $0 \leq P \leq 1$

[Go to Discussion \(https://www.test.aspirestudy.in/library/qus/10201\)](https://www.test.aspirestudy.in/library/qus/10201)

nimcet Previous Year PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=sub&key=Previous Year>)

nimcet NIMCET 2015 PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=topic&key=NIMCET 2015>)

Qus : 53 nimcet PYQ 2015 (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=year&key=2015>)

If a, b, c are in geometric progression, then $\log_{ax}^a, \log_{bx}^a$ and $\log_{cx}^a$ are in

- 1 Arithmetic progression
- 2 Geometric progression
- 3 Harmonic progression
- 4 Arithmetico-geometric progression

[Go to Discussion \(https://www.test.aspirestudy.in/library/qus/10202\)](https://www.test.aspirestudy.in/library/qus/10202)

nimcet Previous Year PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=sub&key=Previous Year>)

nimcet NIMCET 2015 PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=topic&key=NIMCET 2015>)

Qus : 54 nimcet PYQ 2015 (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=year&key=2015>)

The value of the sum $\frac{1}{2\sqrt{1}+1\sqrt{2}} + \frac{1}{3\sqrt{2}+2\sqrt{3}} + \frac{1}{4\sqrt{3}+3\sqrt{4}} + \dots + \frac{1}{25\sqrt{24}+24\sqrt{25}}$ is

- ☐ 1 $\frac{9}{10}$
- ☐ 2 $\frac{4}{5}$
- ☐ 3 $\frac{14}{15}$
- ☐ 4 $\frac{7}{15}$

[Go to Discussion \(https://www.test.aspirestudy.in/library/qus/10203\)](https://www.test.aspirestudy.in/library/qus/10203)

nimcet Previous Year PYQ ([https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=sub&key=Previous Year](https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=sub&key=Previous%20Year))

nimcet NIMCET 2015 PYQ ([https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=topic&key=NIMCET 2015](https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=topic&key=NIMCET%2015))

Qus : 55 nimcet PYQ 2015 (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=year&key=2015>)

If $\vec{a} = \hat{i} - \hat{k}$, $\vec{b} = x\hat{i} + \hat{j} + (1 - x)\hat{k}$ and $\vec{c} = y\hat{i} + x\hat{j} + (1 + x - y)\hat{k}$, then $[\vec{a}, \vec{b}, \vec{c}]$ depends on

- ☐ 1 Neither x nor y
- ☐ 2 Only x
- ☐ 3 Only y
- ☐ 4 Both x and y

[Go to Discussion \(https://www.test.aspirestudy.in/library/qus/10204\)](https://www.test.aspirestudy.in/library/qus/10204)

nimcet Previous Year PYQ ([https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=sub&key=Previous Year](https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=sub&key=Previous%20Year))

nimcet NIMCET 2015 PYQ ([https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=topic&key=NIMCET 2015](https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=topic&key=NIMCET%2015))

Qus : 56 nimcet PYQ 2015 (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=year&key=2015>)

If $42({}^nP_2) = ({}^nP_4)$ then the value of n is

- 1
- 2
- 3
- 4

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nimcet Previous Year PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=sub&key=Previous Year>)

nimcet NIMCET 2015 PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=topic&key=NIMCET 2015>)

Qus : 57 nimcet PYQ 2015 (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=year&key=2015>)

The foot of the perpendicular from the point (2, 4) upon $x + y = 1$ is

- 1
- 2
- 3
- 4

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nimcet Previous Year PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=sub&key=Previous Year>)

nimcet NIMCET 2015 PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=topic&key=NIMCET 2015>)

Qus : 58 nimcet PYQ 2015 (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=year&key=2015>)

The value of k for which the equation $(k - 2)x^2 + 8x + k + 4 = 0$ has both real, distinct and negative roots is

- 1

2

2

3

3

4

-4

[Go to Discussion \(https://www.test.aspirestudy.in/library/qus/10207\)](https://www.test.aspirestudy.in/library/qus/10207)

nimcet Previous Year PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=sub&key=Previous Year>)

nimcet NIMCET 2015 PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=topic&key=NIMCET 2015>)

Qus : 59 nimcet PYQ 2015 (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=year&key=2015>)

If $(2, 1)$, $(-1, -2)$, $(3, 3)$ are the midpoints of the sides BC, CA, AB of a triangle ABC, then equation of the line BC is

1

$$5x + 4y + 6 = 0$$

2

$$5x - 4y - 6 = 0$$

3

$$5x + 4y - 6 = 0$$

4

$$5x - 4y + 6 = 0$$

[Go to Discussion \(https://www.test.aspirestudy.in/library/qus/10208\)](https://www.test.aspirestudy.in/library/qus/10208)

nimcet Previous Year PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=sub&key=Previous Year>)

nimcet NIMCET 2015 PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=topic&key=NIMCET 2015>)

Qus : 60 nimcet PYQ 2015 (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=year&key=2015>)

If a fair dice is rolled successively, then the probability that 1 appears in an even numbered throw is

1

$$\frac{5}{36}$$

2

$$\frac{6}{11}$$

3

$$\frac{1}{6}$$

4

$$\frac{5}{11}$$

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nimcet Previous Year PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=sub&key=Previous Year>)

nimcet NIMCET 2015 PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=topic&key=NIMCET 2015>)

Qus : 61 nimcet PYQ 2015 (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=year&key=2015>)

Let $\vec{a} = \hat{i} + \hat{j} + \hat{k}$, $\vec{b} = \hat{i} - \hat{j} + \hat{k}$ and $\vec{c} = \hat{i} - \hat{j} - \hat{k}$ be three vectors. A vector $\vec{v}$ in the plane of $\vec{a}$ and $\vec{b}$ whose projection on $\frac{\vec{c}}{|\vec{c}|}$ is $\frac{1}{\sqrt{3}}$, is

1

$$3\hat{i} - \hat{j} + 3\hat{k}$$

2

$$\hat{i} - 3\hat{j} + 3\hat{k}$$

3

$$5\hat{i} - 2\hat{j} + 5\hat{k}$$

4

$$2\hat{i} - \hat{j} + 3\hat{k}$$

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nimcet Previous Year PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=sub&key=Previous Year>)

nimcet NIMCET 2015 PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=topic&key=NIMCET 2015>)

Qus : 62 nimcet PYQ 2015 (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=year&key=2015>)

The number of bit strings of length 10 that contain either five consecutive 0's or five consecutive 1's is

1

64

2

112

3

220

4

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nimcet Previous Year PYQ

(<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=sub&key=Previous Year>)

nimcet NIMCET 2015 PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=topic&key=NIMCET 2015>)

Qus : 63 nimcet PYQ 2015 (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=year&key=2015>)

If $0 < x < \pi$ and $\cos x + \sin x = \frac{1}{2}$, then the value of $\tan x$ is

1

$$\frac{4-\sqrt{7}}{3}$$

2

$$\frac{4+\sqrt{7}}{3}$$

3

$$\frac{1+\sqrt{7}}{4}$$

4

$$\frac{1-\sqrt{7}}{4}$$

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nimcet Previous Year PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=sub&key=Previous Year>)

nimcet NIMCET 2015 PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=topic&key=NIMCET 2015>)

Qus : 64 nimcet PYQ 2015 (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=year&key=2015>)

If $\vec{a}$, $\vec{b}$ and $\vec{c}$ are the position vectors of the vertices A, B, C of a triangle ABC, then the area of the triangle ABC is

1

$$\frac{1}{2} |\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}|$$

2

$$|\vec{a} \times \vec{b}|$$

3

$$\frac{1}{2} |\vec{a} \times \vec{b} - \vec{b} \times \vec{c} - \vec{c} \times \vec{a}|$$

4

$$\vec{a} \times (\vec{b} \times \vec{c})$$

[Go to Discussion \(https://www.test.aspirestudy.in/library/qus/10213\)](https://www.test.aspirestudy.in/library/qus/10213)

nimcet Previous Year PYQ

(<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=sub&key=Previous Year>)

nimcet NIMCET 2015 PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=topic&key=NIMCET 2015>)

Qus : 65 nimcet PYQ 2015 (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=year&key=2015>)

If $\int e^x(f(x) - f'(x))dx = \phi(x)$, then the value of $\int e^x f(x)dx$ is

1

$\phi(x) + e^x f(x)$

2

$\phi(x) - e^x f(x)$

3

$\frac{1}{2}[e^x f(x) + \phi(x)]$

4

$\frac{1}{2}[e^x f'(x) + \phi(x)]$

[Go to Discussion \(https://www.test.aspirestudy.in/library/qus/10214\)](https://www.test.aspirestudy.in/library/qus/10214)

nimcet Previous Year PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=sub&key=Previous Year>)

nimcet NIMCET 2015 PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=topic&key=NIMCET 2015>)

Qus : 66 nimcet PYQ 2015 (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=year&key=2015>)

If $3x + 4y + k = 0$ is a tangent to the hyperbola $9x^2 - 16y^2 = 144$ then the value of K is

1

0

2

1

3

-1

4

-3

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nimcet Previous Year PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=sub&key=Previous Year>)

nimcet NIMCET 2015 PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=topic&key=NIMCET 2015>)

Qus : 67 nimcet PYQ 2015 (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=year&key=2015>)

a, b, c are positive integers such that $a^2 + 2b^2 - 2bc = 100$ and $2ab - c^2 = 100$. Then the value of $\frac{a+b}{c}$ is

1 10

2 100

3 2

4 20

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nimcet Previous Year PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=sub&key=Previous Year>)

nimcet NIMCET 2015 PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=topic&key=NIMCET 2015>)

Qus : 68 nimcet PYQ 2015 (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=year&key=2015>)

If $(-4, 5)$ is one vertex and $7x - y + 8 = 0$ is one diagonal of a square, then the equation of the other diagonal is

1 $x + 7y = 21$

2 $x + 7y = 31$

3 $x + 7y = 28$

4 $x + 7y = 35$

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nimcet Previous Year PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=sub&key=Previous Year>)

nimcet NIMCET 2015 PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=topic&key=NIMCET 2015>)

Qus : 69 nimcet PYQ 2015 (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=year&key=2015>)

Out of $2n + 1$ tickets, which are consecutively numbered, three are drawn at random. Then the probability that the numbers on them are in arithmetic progression is

- 1 ☐ $\frac{n^2}{4n^2-1}$
- 2 ☐ $\frac{n}{4n^2-1}$
- 3 ☐ $\frac{3n}{4n^2-1}$
- 4 ☐ $\frac{3n^2}{4n^2-1}$

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nimcet Previous Year PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=sub&key=Previous Year>)

nimcet NIMCET 2015 PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=topic&key=NIMCET 2015>)

Qus : 70 nimcet PYQ 2015 (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=year&key=2015>)

A circle touches the X-axis and also touches another circle with centre at $(0, 3)$ and radius 2. Then the locus of the centre of the first circle is

- 1 ☐ a parabola
- 2 ☐ a hyperbola
- 3 ☐ a circle
- 4 ☐ an ellipse

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nimcet Previous Year PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=sub&key=Previous Year>)

nimcet NIMCET 2015 PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=topic&key=NIMCET 2015>)

Qus : 71 nimcet PYQ 2015 (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=year&key=2015>)

Let $\bar{P}$ and $\bar{Q}$ denote the complements of two sets P and Q. Then the set $(P - Q) \cup (Q - P) \cup (P \cap Q)$ is

1

$$P \cup Q$$

2

$$\bar{P} \cup \bar{Q}$$

3

$$P \cap Q$$

4

$$\bar{P} \cap \bar{Q}$$

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nimcet Previous Year PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=sub&key=Previous Year>)

nimcet NIMCET 2015 PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=topic&key=NIMCET 2015>)

Qus : 72 nimcet PYQ 2015 (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=year&key=2015>)

With the usual notation $\frac{d^2x}{dy^2}$

1

$$\left(\frac{d^2y}{dx^2}\right)^{-1}$$

2

$$\frac{d^2y}{dx^2} \left(\frac{dy}{dx}\right)^2$$

3

$$-\left(\frac{d^2y}{dx^2}\right)^{-1} \left(\frac{dy}{dx}\right)^{-3}$$

4

$$-\left(\frac{d^2y}{dx^2}\right) \left(\frac{dy}{dx}\right)^{-3}$$

[Go to Discussion \(https://www.test.aspirestudy.in/library/qus/10221\)](https://www.test.aspirestudy.in/library/qus/10221)

nimcet Previous Year PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=sub&key=Previous Year>)

nimcet NIMCET 2015 PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=topic&key=NIMCET 2015>)

Qus : 73 nimcet PYQ 2015 (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=year&key=2015>)

The radius of the circle passing through the foci of the ellipse $\frac{x^2}{16} + \frac{y^2}{9}$ and having its centre at (0, 3) is

1 4 units

2 3 units

3 $\sqrt{12}$ units

4 $\frac{7}{2}$ units

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nimcet Previous Year PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=sub&key=Previous Year>)

nimcet NIMCET 2015 PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=topic&key=NIMCET 2015>)

Qus : 74 nimcet PYQ 2015 (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=year&key=2015>)

A function $f : (0, \pi) \rightarrow \mathbb{R}$ defined by $f(x) = 2\sin x + \cos 2x$ has

1 A local minimum but no local maximum

2 A local maximum but no local minimum

3 Both local minimum and local maximum

4 Neither a local minimum nor a local maximum

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nimcet Previous Year PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=sub&key=Previous Year>)

nimcet NIMCET 2015 PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=topic&key=NIMCET 2015>)

Qus : 75 nimcet PYQ 2015 (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=year&key=2015>)

A matrix M_r is defined as $M_r = \begin{bmatrix} r & r-1 \\ r-1 & r \end{bmatrix}$, $r \in N$ then the value of $\det(M_1) + \det(M_2) + \dots + \det(M_{2015})$ is

- ☐ 1
- ☐ 2
- ☐ 3
- ☐ 4

[Go to Discussion \(https://www.test.aspirestudy.in/library/qus/10224\)](https://www.test.aspirestudy.in/library/qus/10224)

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nimcet NIMCET 2015 PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=topic&key=NIMCET 2015>)

Qus : 76 nimcet PYQ 2015 (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=year&key=2015>)

If $\vec{AC} = 2\hat{i} + \hat{j} + \hat{k}$ and $\vec{BD} = -\hat{i} + 3\hat{j} + 2\hat{k}$ then the area of the quadrilateral ABCD is

- ☐ 1
- ☐ 2
- ☐ 3
- ☐ 4

[Go to Discussion \(https://www.test.aspirestudy.in/library/qus/10225\)](https://www.test.aspirestudy.in/library/qus/10225)

nimcet Previous Year PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=sub&key=Previous Year>)

nimcet NIMCET 2015 PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=topic&key=NIMCET 2015>)

Qus : 77 nimcet PYQ 2015 (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=year&key=2015>)

The value of $\sin^{-1} \frac{1}{\sqrt{2}} + \sin^{-1} \frac{\sqrt{2}-\sqrt{1}}{\sqrt{6}} + \sin^{-1} \frac{\sqrt{3}-\sqrt{2}}{\sqrt{12}} + \dots$ to infinity, is equal to

- 1
- 2
- 3
- 4

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nimcet NIMCET 2015 PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=topic&key=NIMCET 2015>)

Qus : 78 nimcet PYQ 2015 (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=year&key=2015>)

If two circles $x^2 + y^2 + 2gx + 2fy = 0$ and $x^2 + y^2 + 2g'x + 2f'y = 0$ touch each other then which of the following is true?

- 1
- 2
- 3
- 4

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nimcet Previous Year PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=sub&key=Previous Year>)

nimcet NIMCET 2015 PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=topic&key=NIMCET 2015>)

Qus : 79 nimcet PYQ 2015 (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=year&key=2015>)

$\int_0^\pi [\cot x] dx$ where $[.]$ denotes the greatest integer function, is equal to

- 1

2

1

3

-1

4

 $\frac{-\pi}{2}$ [Go to Discussion \(https://www.test.aspirestudy.in/library/qus/10228\)](https://www.test.aspirestudy.in/library/qus/10228)

nimcet Previous Year PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=sub&key=Previous Year>)

nimcet NIMCET 2015 PYQ (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=topic&key=NIMCET 2015>)

Qus : 80 nimcet PYQ 2015 (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=year&key=2015>)

In a right angled triangle, the hypotenuse is four times the perpendicular drawn to it from the opposite vertex. The value of one of the acute angles is

1

 45°

2

 30°

3

 15°

4

None of these

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A is targeting B, B and C are targeting A. Probability of hitting the target by A, B and C are $\frac{2}{3}$, $\frac{1}{2}$ and $\frac{1}{3}$ respectively. If A is hit then the probability that B hits the target and C does not, is

1

 $\frac{1}{2}$

2

 $\frac{1}{3}$

3

 $\frac{2}{3}$

4

$$\frac{3}{4}$$

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Qus : 82 nimcet PYQ 2015 (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=year&key=2015>)

If the angles of a triangle are in the ratio 2 : 3 : 7, then the ratio of the sides opposite to these angles is

1

$$\sqrt{2} : 2 : \sqrt{3} + 1$$

2

$$2 : \sqrt{2} : \sqrt{3} + 1$$

3

$$2 : \sqrt{2} : \frac{\sqrt{2}}{\sqrt{3} - 1}$$

4

$$\frac{1}{\sqrt{2}} : 2 : \frac{\sqrt{3} + 1}{2}$$

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Suppose that A and B are two events with probabilities $P(A) = \frac{1}{2}$ $P(B) = \frac{1}{3}$ Then which of the following is true?

1

$$1 < P(A \cap B) < 1$$

1 $\frac{1}{3} \leq P(A \cap B) \leq \frac{1}{2}$

2 $\frac{1}{4} \leq P(A \cap B) \leq \frac{1}{3}$

3 $\frac{1}{6} \leq P(A \cap B) \leq \frac{1}{3}$

4 $\frac{1}{4} \leq P(A \cap B) \leq \frac{1}{2}$

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Qus : 84 nimcet PYQ 2015 (<https://www.test.aspirestudy.in/library/pyq/1/nimcet?view=year&key=2015>)

The number of one-to-one functions from $\{1, 2, 3\}$ to $\{1, 2, 3, 4, 5\}$ is

1 125

2 243

3 10

4 60

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A harbour lies in a direction 60° South of West from a fort and at a distance 30 km from it, a ship sets out from the harbour at noon and sails due East at 10 km an hour. The time at which the ship will be 70 km from the fort is

- 1 7 PM
- 2 8 PM
- 3 5 PM
- 4 10 PM

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If x, y, z are three consecutive positive integers, then $\log(1 + xz)$ is

- 1 $\log y$
- 2 $\log \frac{y}{2}$
- 3 $\log 2y$
- 4 $2\log y$

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A professor has 24 text books on computer science and is concerned about their coverage of the topics (P) compilers, (Q) data structures and (R) Operating systems. The following data gives the number of books that contain material on these topics:

$$n(P) = 8, n(Q) = 13, n(R) = 13, n(P \cap R) = 3, n(P \cap Q) = 3, n(Q \cap R) = 3, n(Q \cap R) = 6, n(P \cap Q \cap R) = 2$$

where $n(x)$ is the cardinality of the set x . Then the number of text books that have no material on compilers is

- ☐ 1 4
- ☐ 2 8
- ☐ 3 12
- ☐ 4 16

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The value of $\tan\left(\frac{7\pi}{8}\right)$ is

- ☐ 1 $1 - \sqrt{2}$
- ☐ 2 $1 + \sqrt{2}$
- ☐ 3 $\sqrt{2} + \sqrt{3}$
- ☐ 4 $\sqrt{2} - \sqrt{3}$

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If $\vec{a}$ and $\vec{b}$ are vectors such that $|\vec{a}| = 13$, $|\vec{b}| = 5$ and $\vec{a} \cdot \vec{b} = 60$ then the value of $|\vec{a} \times \vec{b}|$ is

- ☐ 1 625
- ☐ 2 335

2

225

3

45

4

25

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Two towers face each other separated by a distance of 25 meters. As seen from the top of the first tower, the angle of depression of the second tower's base is 60° and that of the top is 30° . The height (in meters) of the second tower is

1

 $\frac{50}{\sqrt{3}}$

2

 $\frac{25}{\sqrt{3}}$

3

50

4

 $25\sqrt{3}$ [Go to Discussion \(https://www.test.aspirestudy.in/library/qus/10239\)](https://www.test.aspirestudy.in/library/qus/10239)

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